



2016 中国互联网络安全大会
China Internet Security Conference

协同联动 共建安全⁺命运共同体

**Can You Trust Autonomous Vehicles:
Contactless Attacks against Sensors of Self-Driving Vehicles**

Who Are We



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安全极客狂欢节



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Roadmap

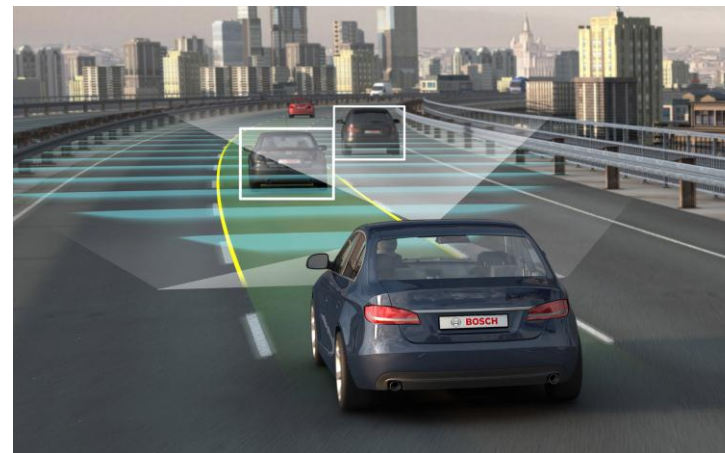


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安全极客狂欢节

- **Autonomous Vehicles**
- **Hacking Sensors**
- **Our Attacks**
 - Ultrasonic sensors
 - MMW radars
 - Cameras
- **Discussion**



The Car Hacking History

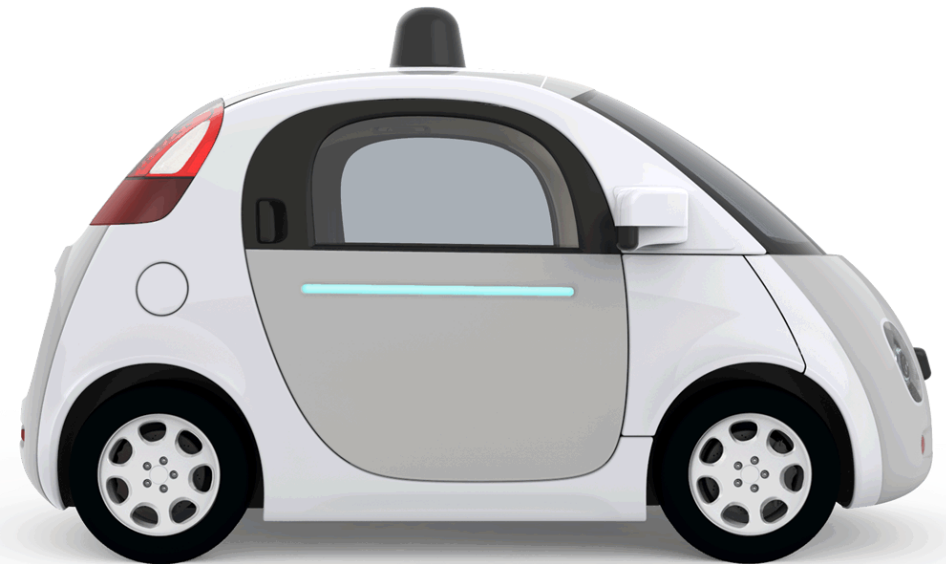


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- **Car ==> CAN bus hacking**
- **Connected car ==> Telematics hacking**
- **Autonomous car ==> Automatic system hacking**



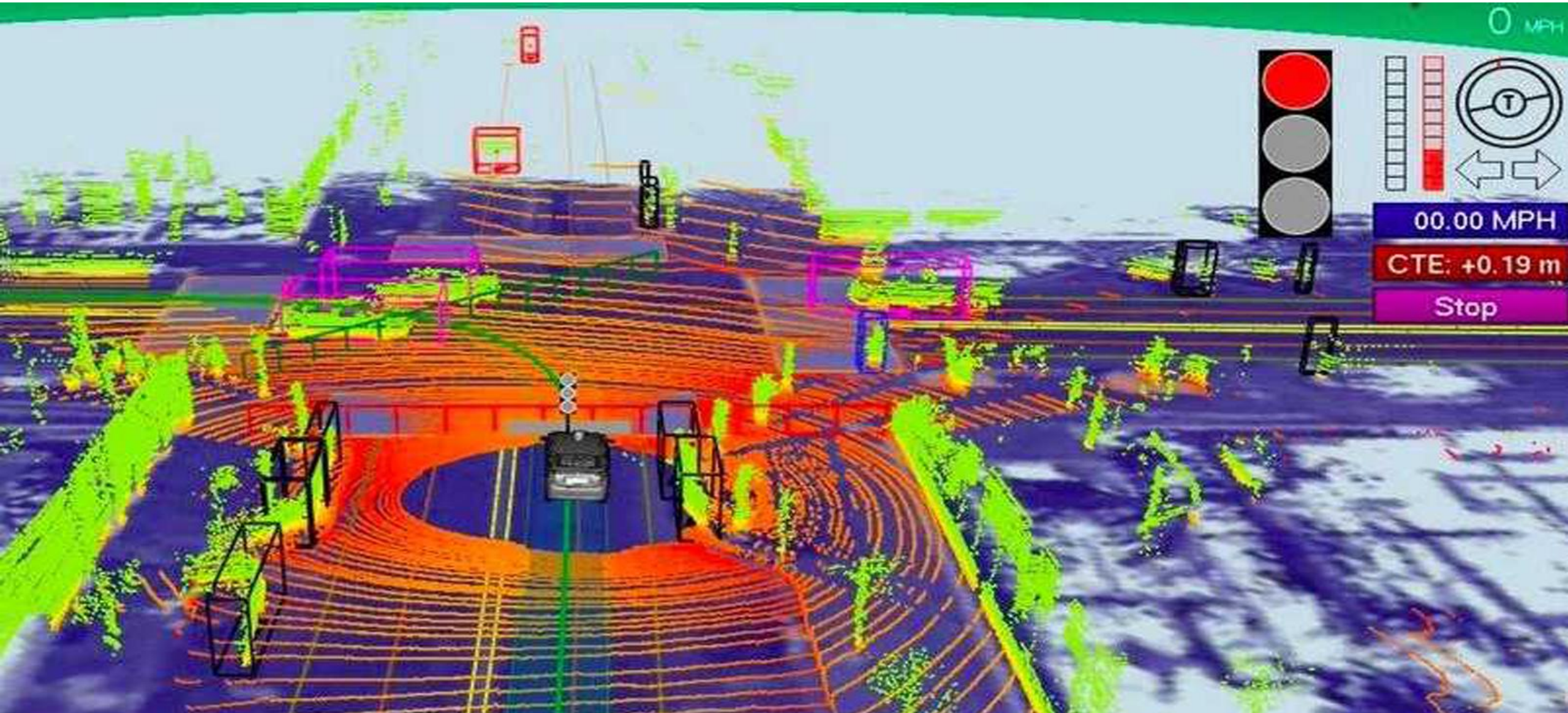
What is Autonomous Vehicle?



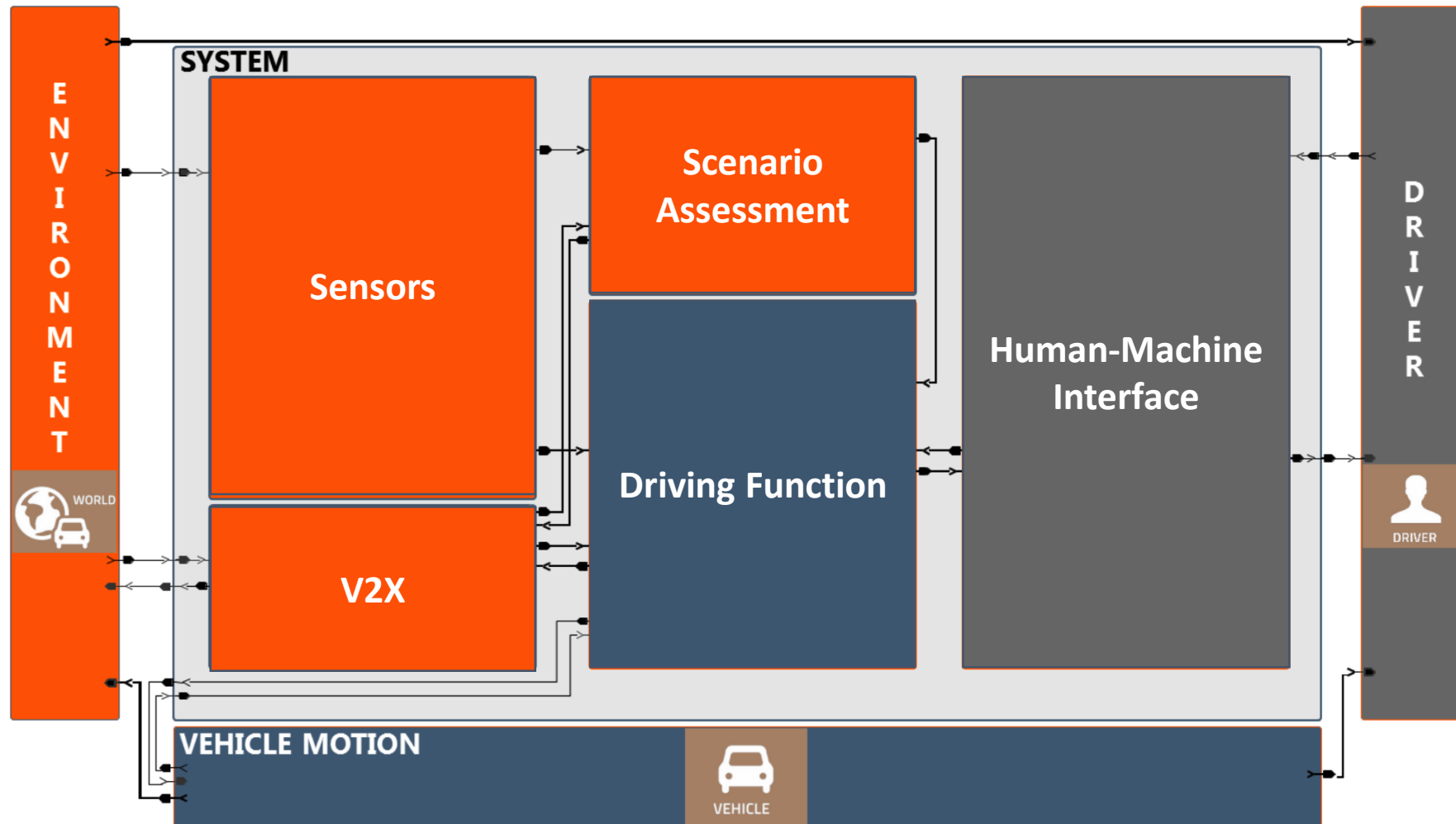
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安全极客狂欢节



Sensors in automated driving system



Sensors for Self-Driving

Cameras

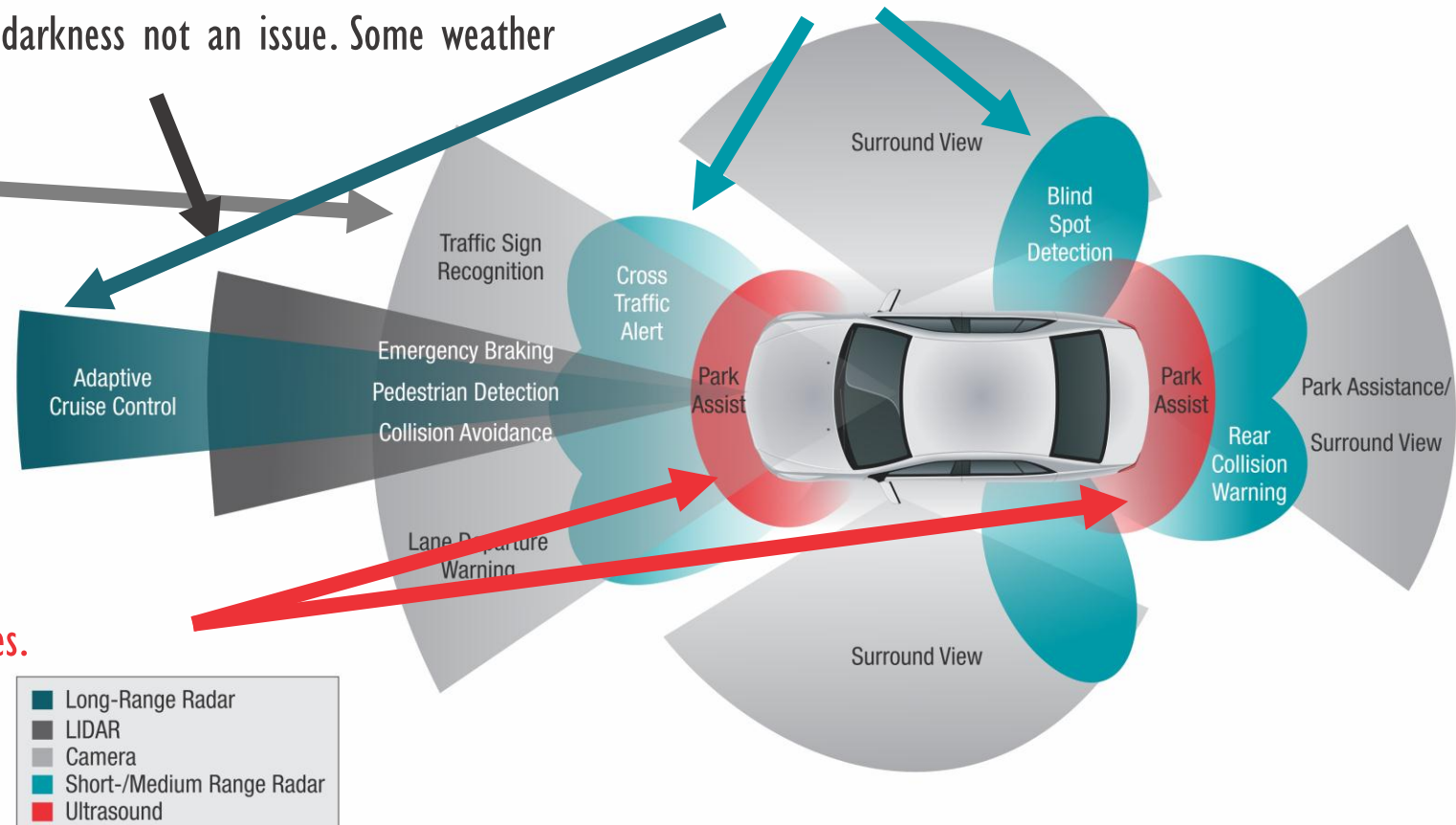
Senses reflected light, limited when dark.
Sees colour, so can be used to read signs, signals, etc.

LiDAR

Emits light, so darkness not an issue. Some weather limitation.

Radar

Works in low light & poor weather, but lower resolution.

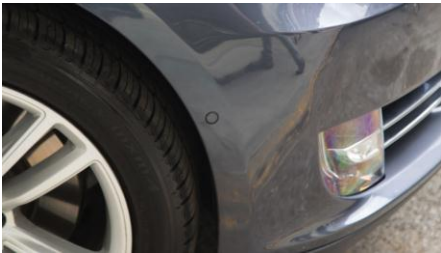


Source: Texas Instruments

How to Hack Sensors

Sensors

Ultrasonic Sensors



Jamming

Spoofing

MMW Radars



Jamming

Spoofing

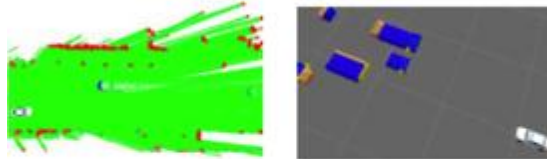
Cameras



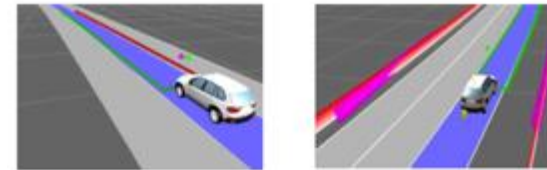
Blinding

Automated System

Representations and Fusion



Road Model and Localization



Situation Interpretation



Control



HMI Display



Tesla: A Tragic Loss



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- **First fatal crash while using Autopilot on May 7, 2016.**
- **Reliability of sensors.**



Source: The New York Times

網易汽车

网易首页 应用

网易考拉

LOFTER

First Tesla Accident in China Caused by Autopilot

国内发生特斯拉第一起自动驾驶事故

2016-08-05 11:21:06 来源: 盖世汽车(上海)

Existing Sensors on Tesla Model S

One MMW Radar

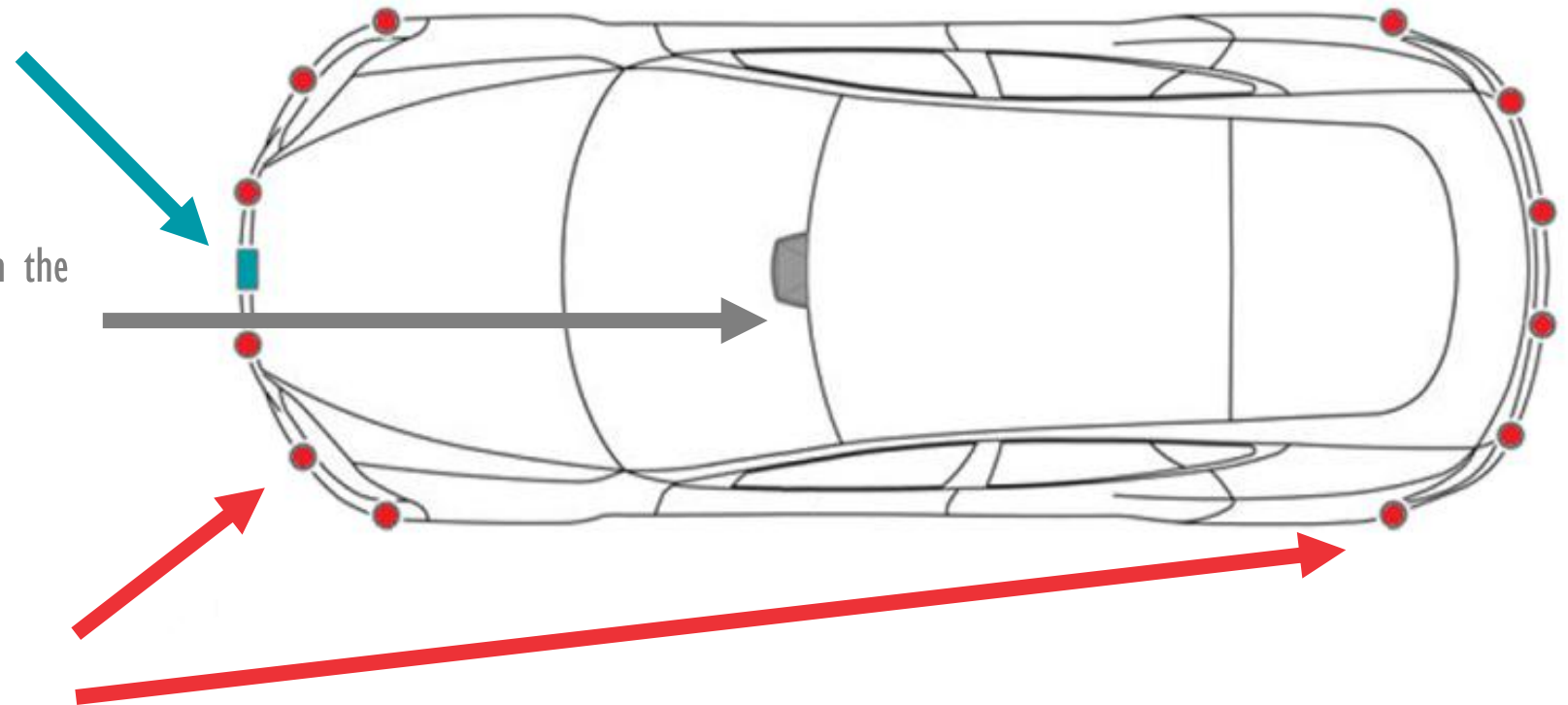
A Medium range Radar is mounted in the front grill.

One camera

A forward looking camera is mounted on the windshield under the rear view mirror.

12 ultrasonic sensors

Ultrasonic sensors are located near the front and rear bumpers.



HMI Display Mistake – Demo on Tesla



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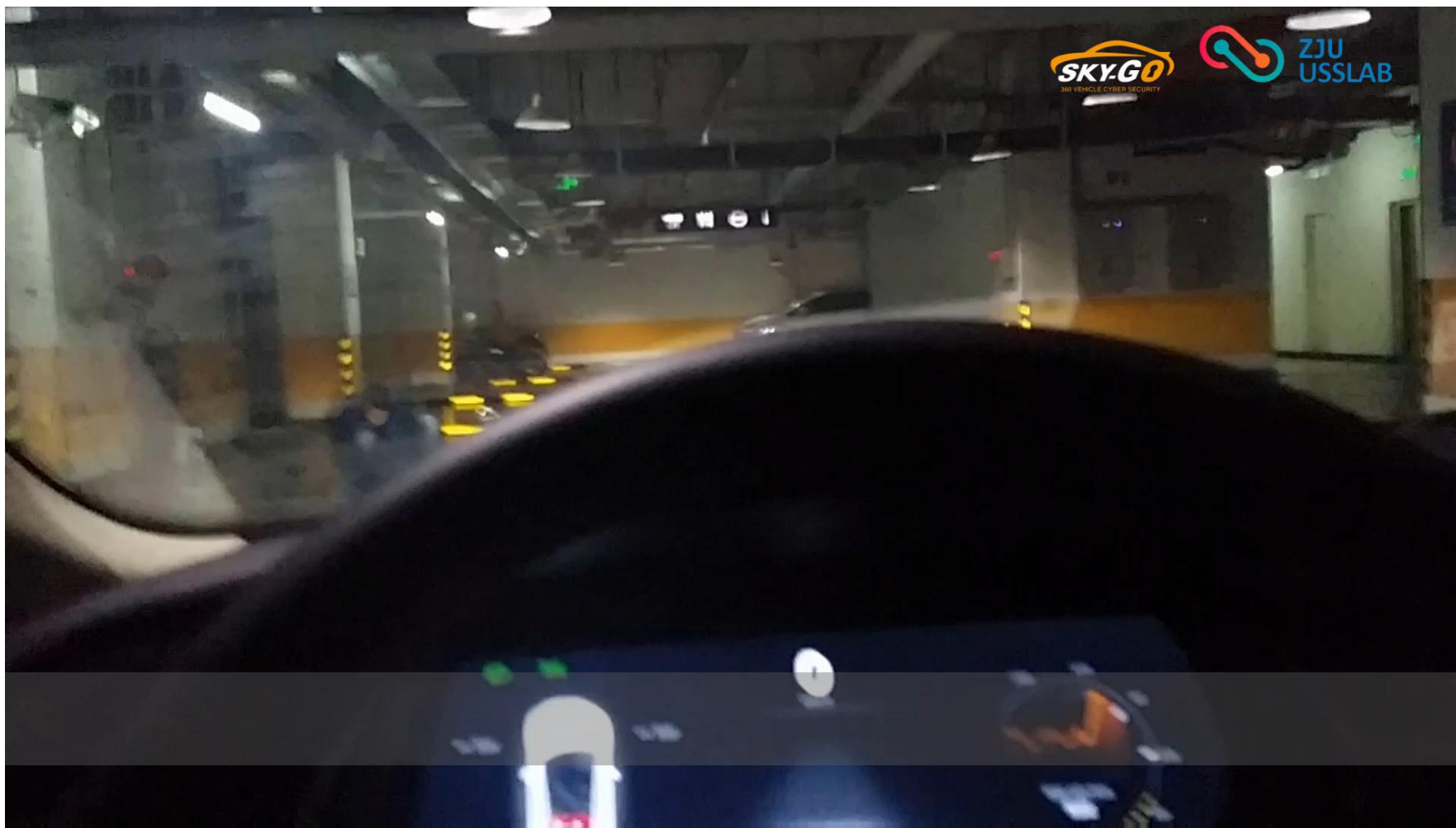
Control Mistake – Demo on Tesla



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Attacking Ultrasonic Sensors

On Tesla, Audi, Volkswagen, and Ford

Ultrasonic Sensor

What is ultrasonic sensor?

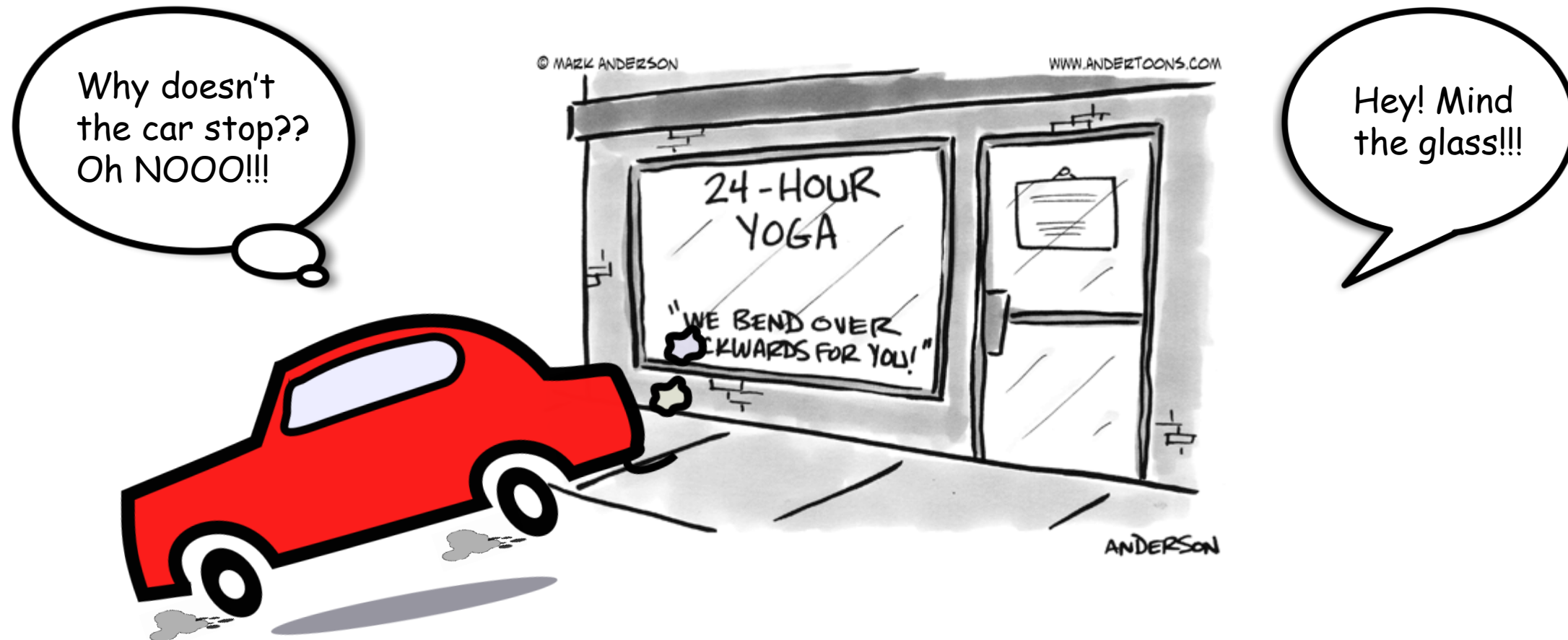
- Measures **distance**
- Proximity sensor ($< 2\text{m}$)

• Applications

- Parking assistance
- Parking space detection
- Self parking
- Tesla's summon

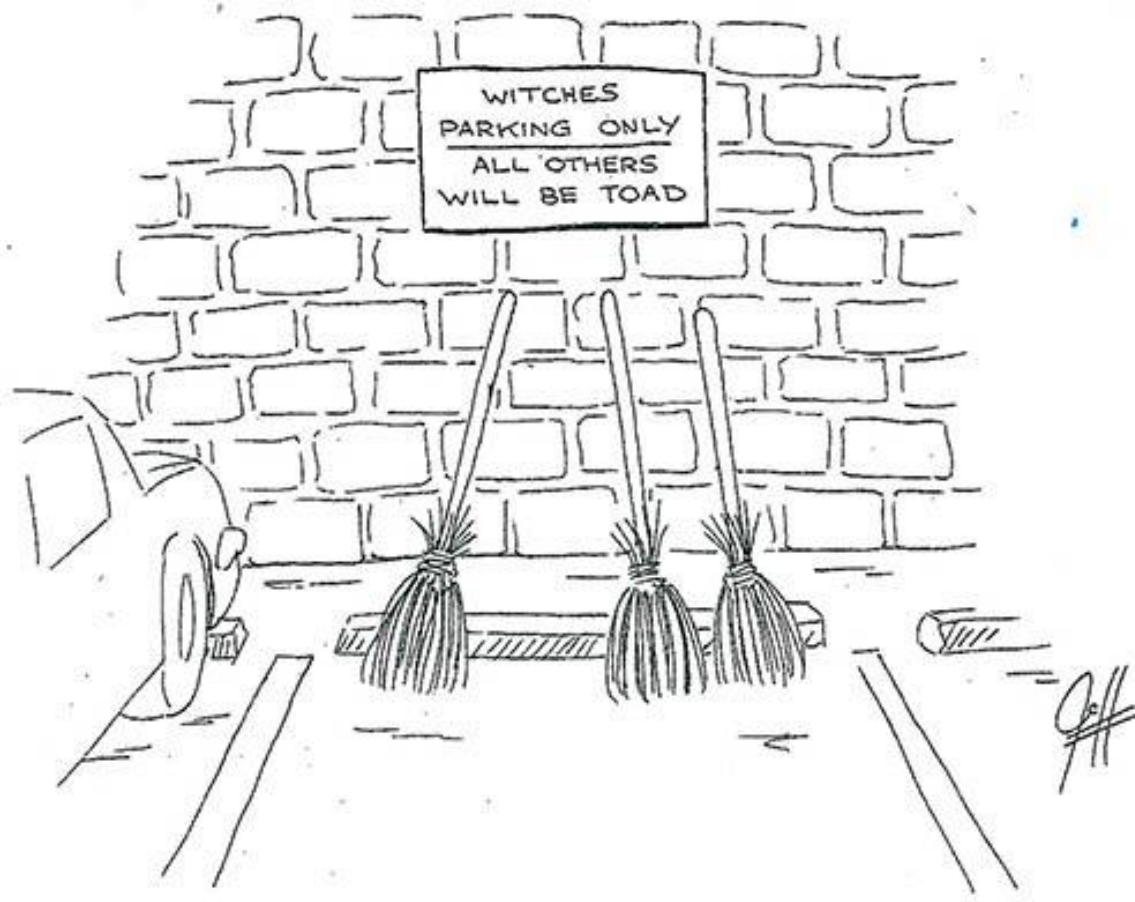


Misuse 1: The car doesn't stop while it should.





Misuse 2: The car stops while it shouldn't.

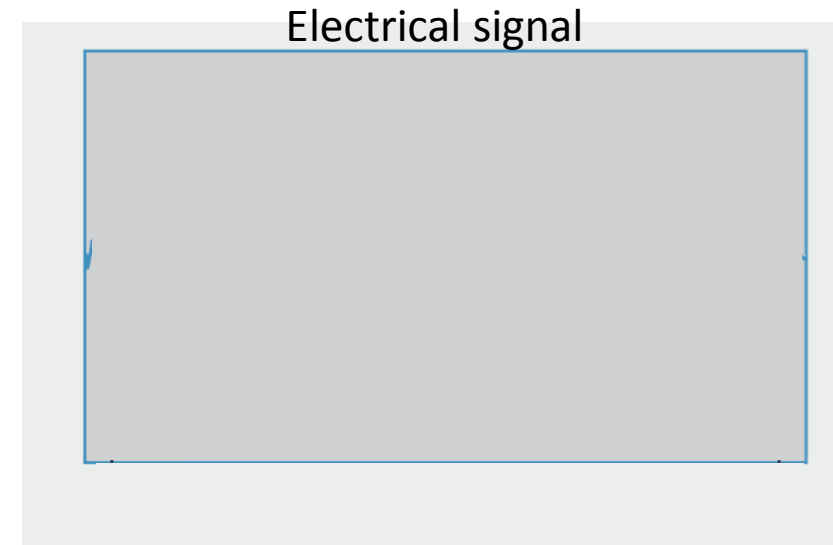
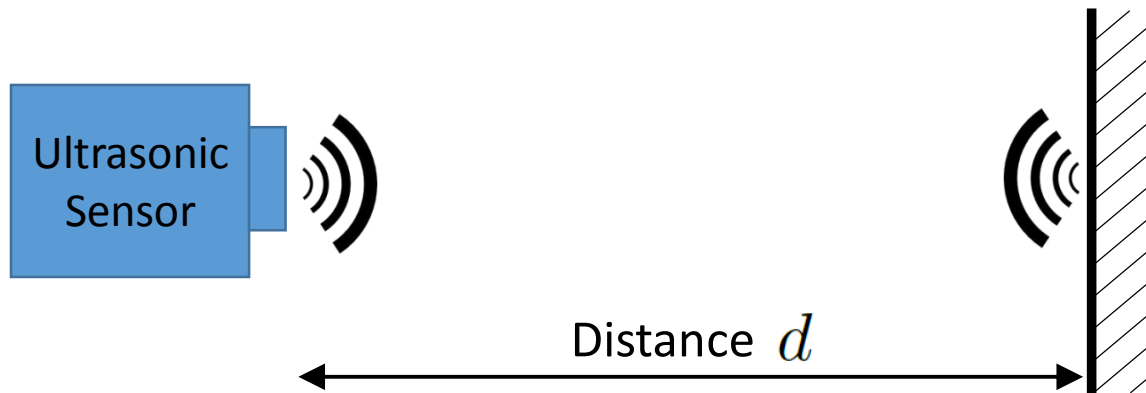


How do ultrasonic sensors work?

- **Emit ultrasound and receive echoes**
- **Piezoelectric Effect**
- **Measure the propagation time (Time of Flight)**
- **Calculate the distance** $d = 0.5 \cdot t_e \cdot c$



t_e : propagation time of echoes
 c : velocity of sound in air



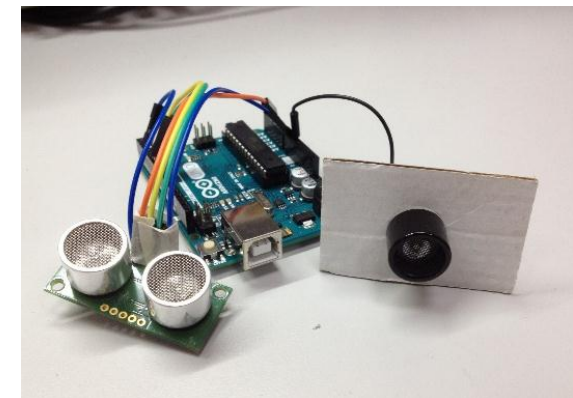
Attacking ultrasonic sensors

Attacks:

- **Jamming** – generates ultrasonic noises – **denial of service**
- **Spoofing** – crafts fake ultrasonic echo pulses – **alters distance**
- **Quieting** – diminishes original ultrasonic echoes – **hides obstacles**

Equipment:

- **Ultrasonic transducers (\$0.4)** – emit ultrasound
- **Signal suppliers – generate excitation signals**
 - Arduino (\$24.95)
 - Signal generator (~\$20)



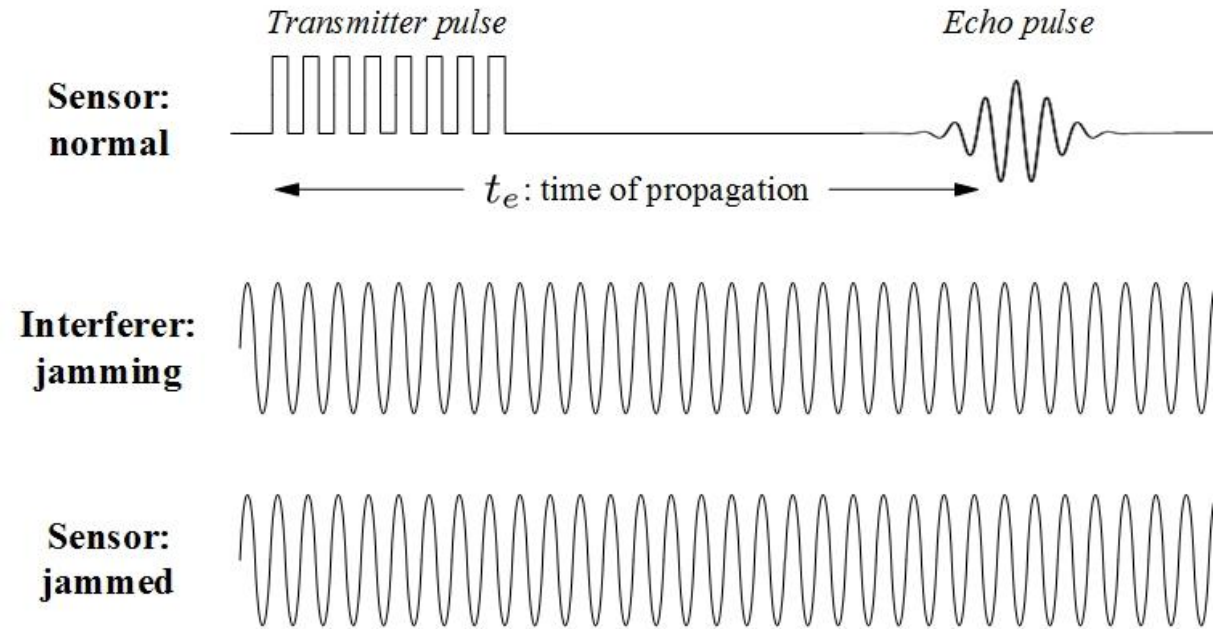
Jamming Attack

- **Basic Idea:**

- Injecting **ultrasonic noises**
- At resonant frequency (40 — 50 kHz)
- Causing **Denial of Service**

- **Tested ultrasonic sensors:**

- In laboratories: 8 models of stand-alone ultrasonic sensors
- Outdoors: Tesla, Audi, Volkswagen, Ford



Jamming Attack – in lab

- **8 models of ultrasonic sensors**

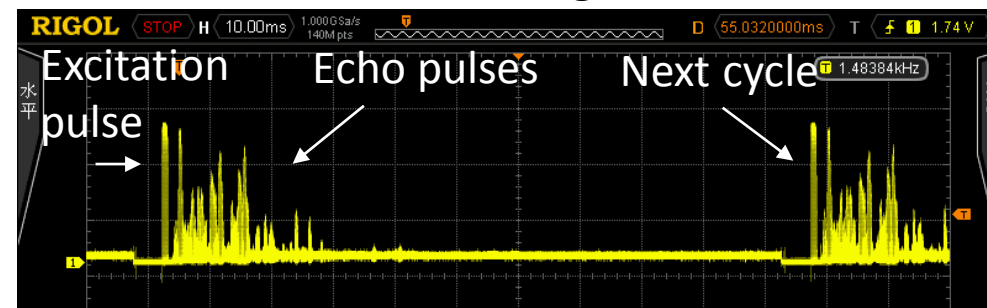
- HC-SR04
- SRF01
- SRF05
- MaxSonar MB1200
- JSN-SR04T
- FreeCars V4
- Grove ultrasonic ranger
- Audi Q3 sensors

- **Sensor reading**

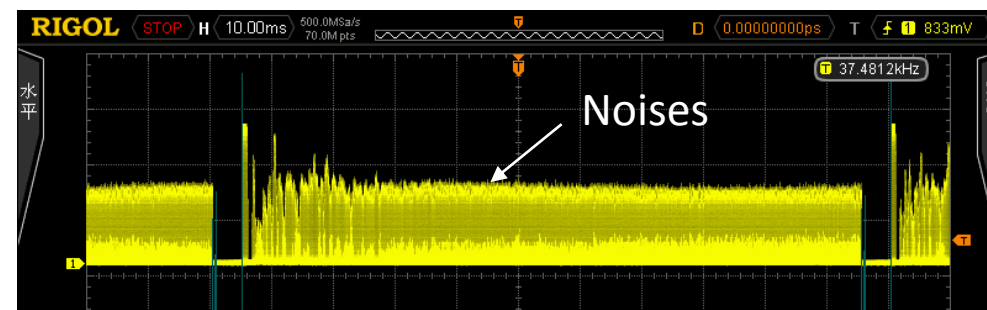
- **Zero** distance
- **Maximum** distance

No jamming

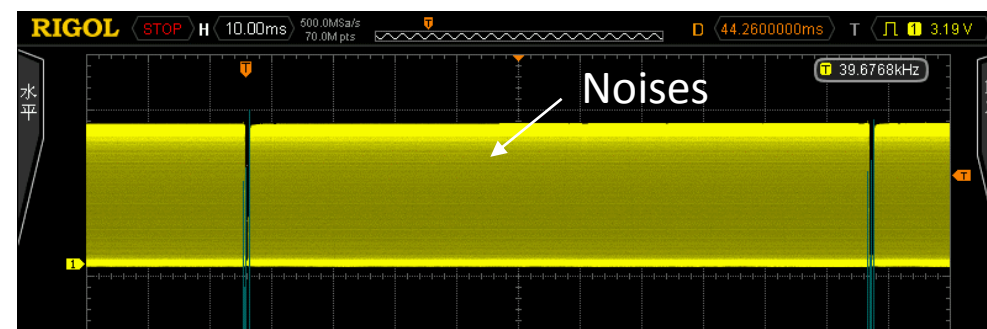
Received electrical signals at the sensor



Weak
Jamming



Strong
Jamming



How should cars behave to jamming?

Zero distance?

or

Maximum distance?

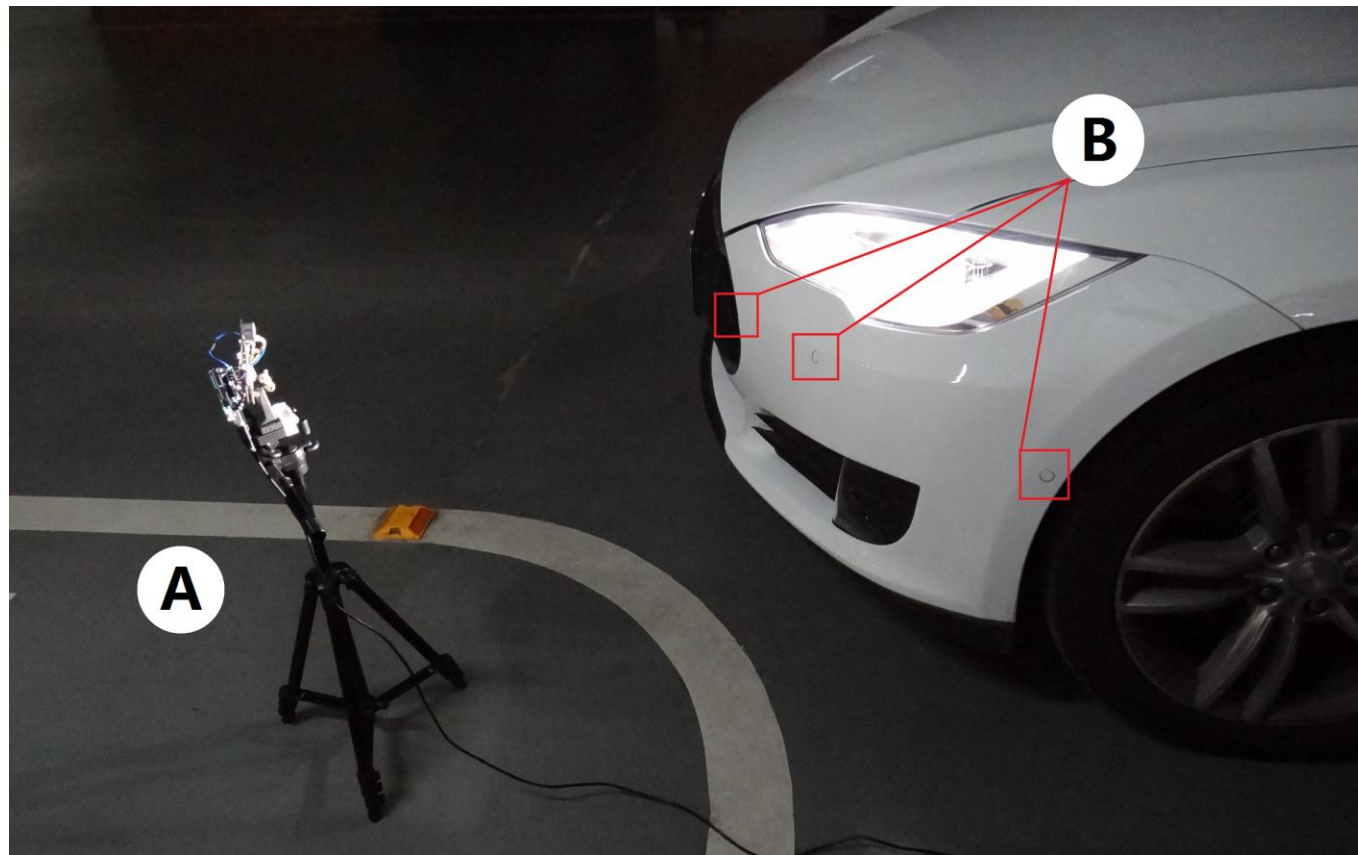
Jamming Attack – on vehicles

- **4 different vehicles**

- Audi Q3
- Volkswagen Tiguan
- Ford Fiesta
- Tesla Model S
 - Self parking
 - Summon

- **Results**

- **Maximum** distance



Experiment setup on Tesla Model S

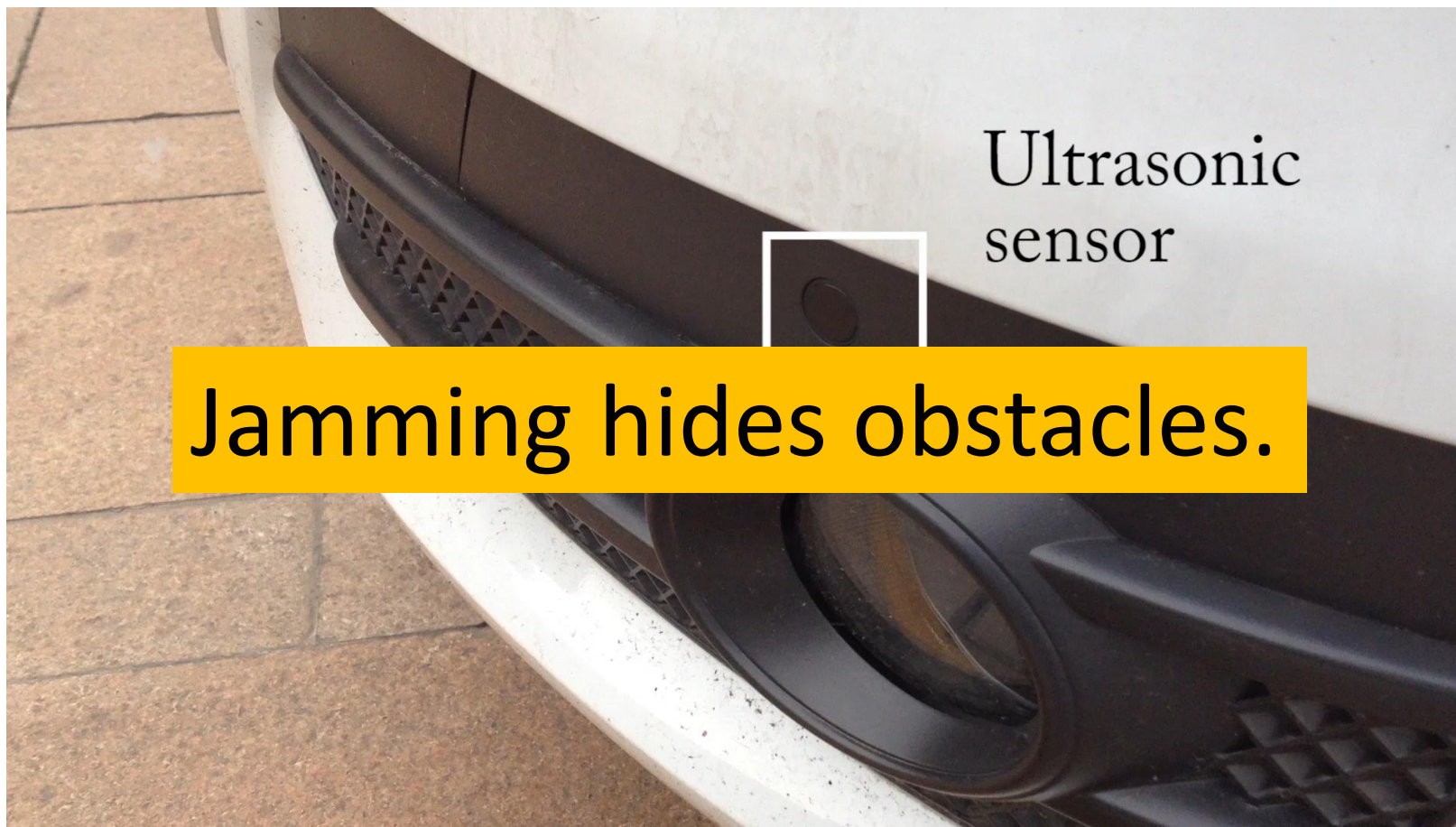
Jamming Attack – Demo on Audi



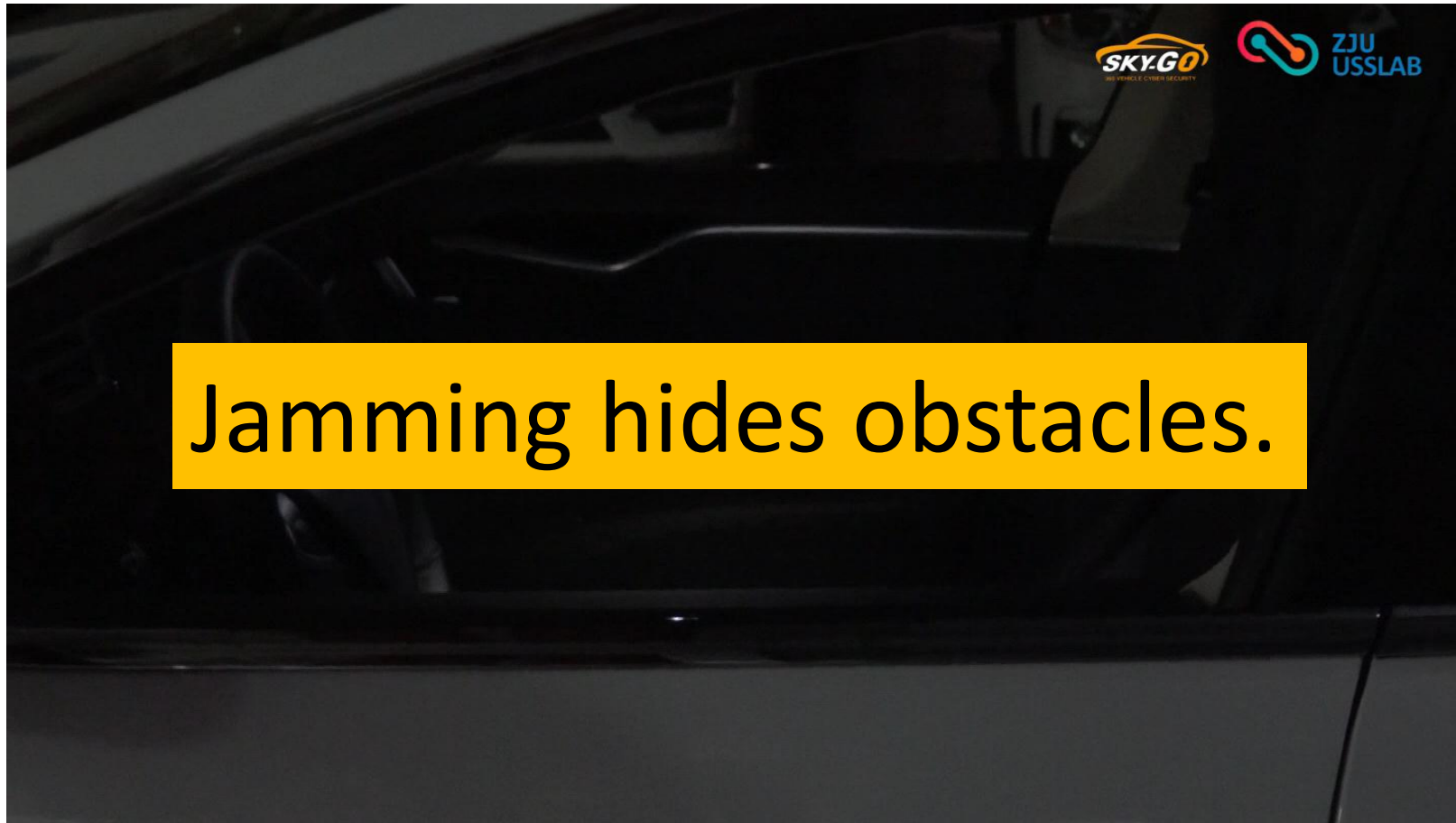
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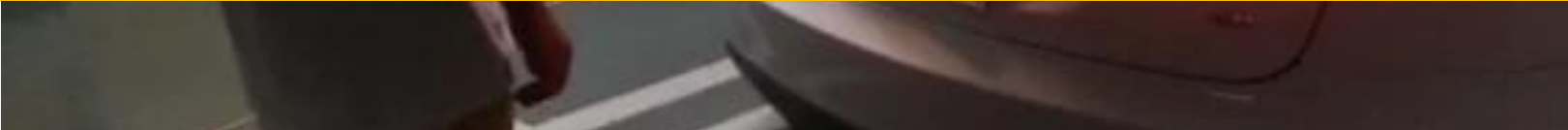
Jamming Attack – Demo on Tesla Summon



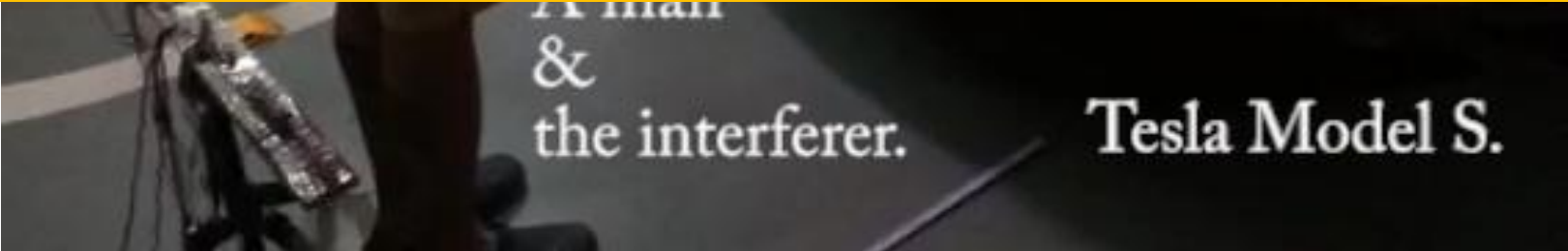
Jamming Attack – Demo on Tesla Summon



The interferer was hit & stopped working.



Jamming distance can be increased.



the interferer.

Tesla Model S.

Jamming Attack – Results

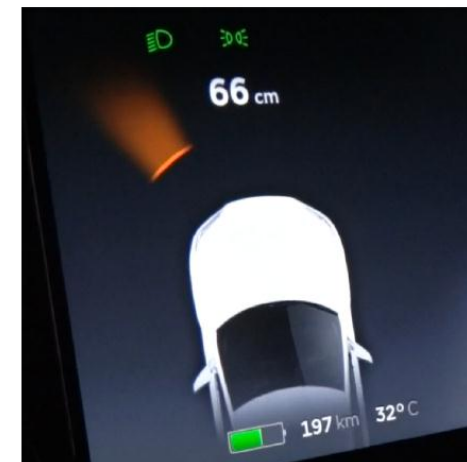
- **On ultrasonic sensors**
 - Zero or maximum distance
- **On vehicles with parking assistance**
 - Maximum distance
- **On self-parking and summon**
 - Car **does not stop** under strong jamming!



Audi Normal



Audi Jammed



Tesla Normal



Tesla Jammed

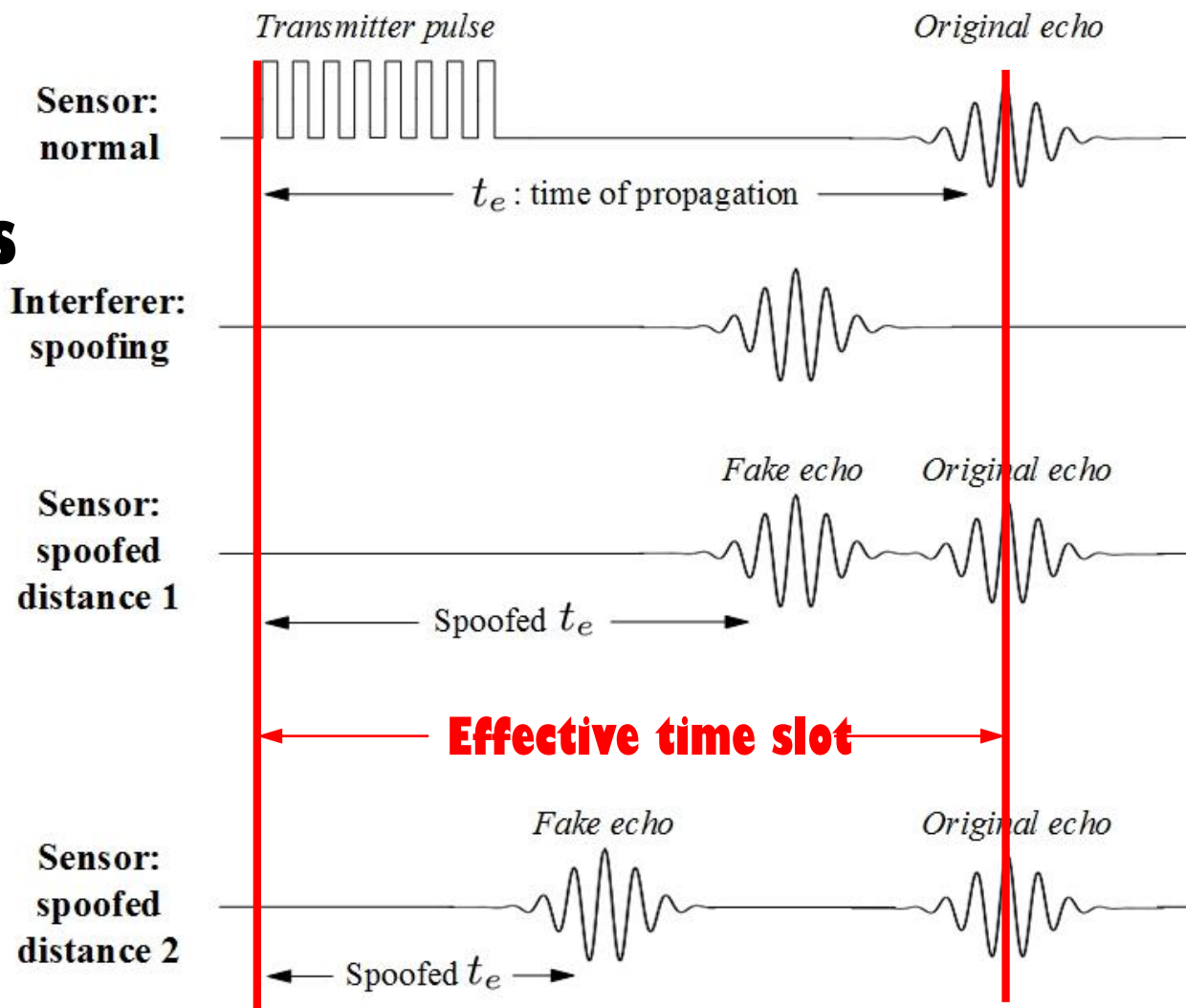
Spoofing Attack

Basic Idea

- Injecting ultrasonic pulses
- At certain time

Non-trivial

- Only the first justifiable echo will be processed
- **Effective time slot**



Spoofing Attack – Demo on Tesla



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Spoofing Attack – Demo on Audi



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安全极客狂欢节



Spoofing Attack – Results



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- **Manipulate sensor readings**

- On stand-alone ultrasonic sensors
- On cars



Tesla Normal



Tesla Spoofed

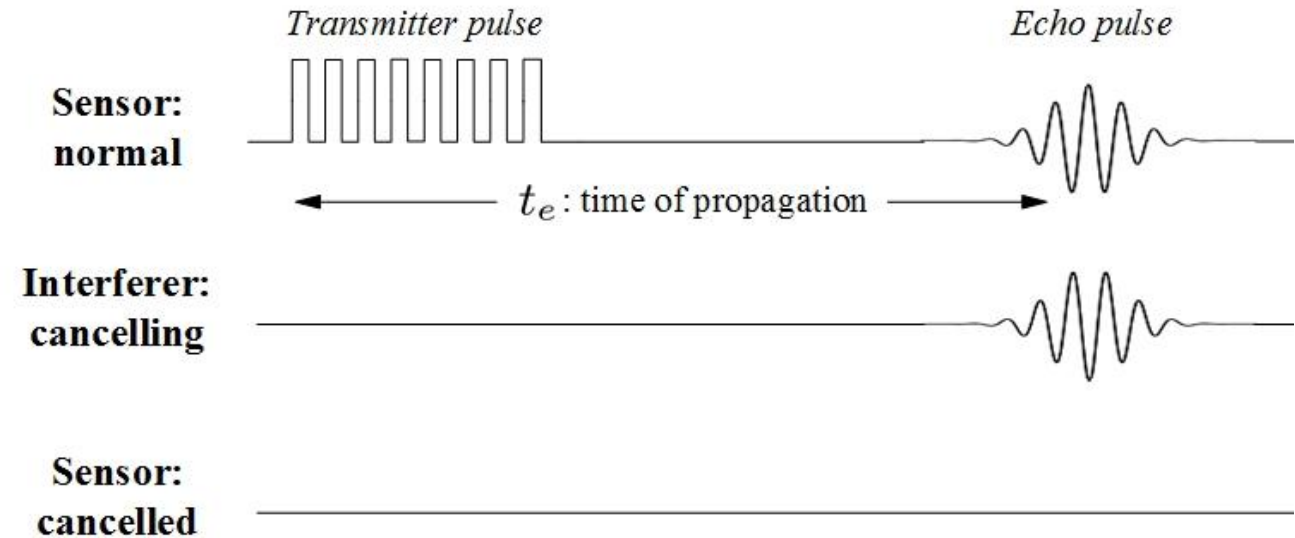


Audi Spoofed

Acoustic Quieting

- **Acoustic Cancellation**

- Cancel original sound with ones of **reversed phase**
- Minor phase and amplitude adjustment



- **Cloaking**

- Sound absorbing materials (e.g., damping foams (\$3/m²))
- Same effect as jamming!



Cloaking Car – Demo



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Attacking Millimeter Wave Radars

On Tesla Model S

Millimeter Wave Radar

What is MMW Radar?

- Measures **distance, angle, speed, shape**
- Short to long range sensing (30-250m)

• Applications

- Adaptive Cruise Control (ACC)
- Collision Avoidance
- Blind Spot Detection



Construction of the Bosch RADAR sensors MRR and LRR3 (Source: Bosch)

Misuse 1: The car doesn't stop while it should.

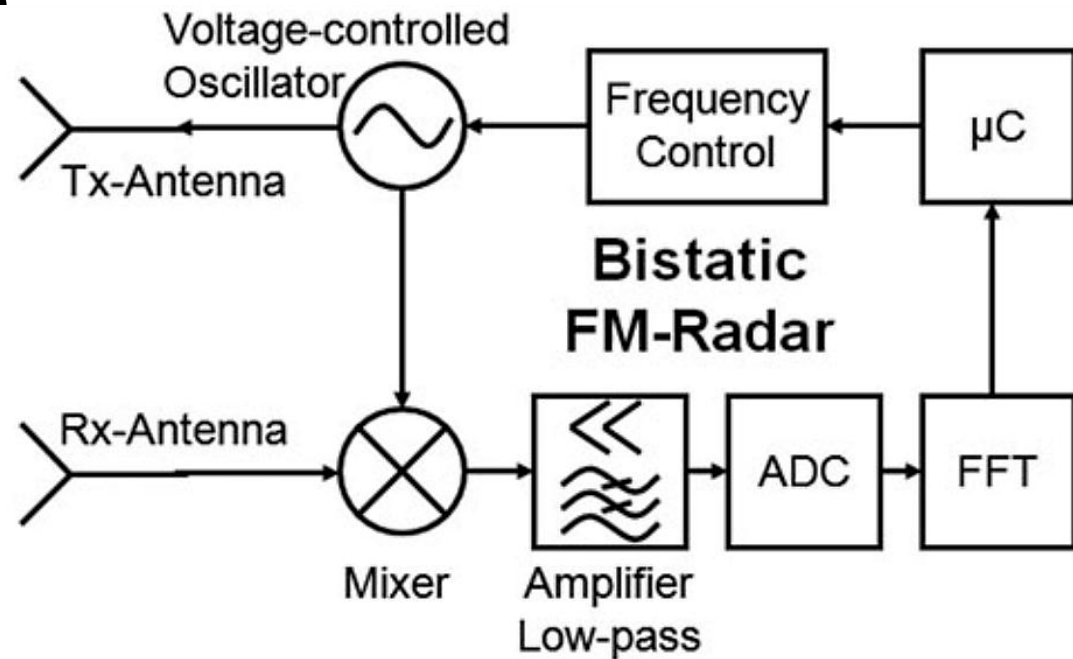


Misuse 2: The car stops while it shouldn't.



How do MMW Radars work?

- **Transmit and receive millimeter electromagnetic waves**
- **Measure the propagation time**
- **Modulation**
 - Amplitude
 - Frequency (FMCW)
 - Phase
- **Doppler Effect**
- **Frequency Bands:**
 - 24 GHz
 - 76-77 GHz



Block diagram of a bistatic Radar with frequency modulation

(Source: H. Winner, Handbook of Driver Assistance Systems)

MMW Radar – To be discovered

#1. Understand Radar signal – Signal Analysis

- Frequency range
- Modulation process
- Ramp height (bandwidth)
- Ramps (number, duration)
- Cycle time

#2. Jamming Attack

- Feasible?
- What jamming signal?

#3. Spoofing Attack

- Feasible?

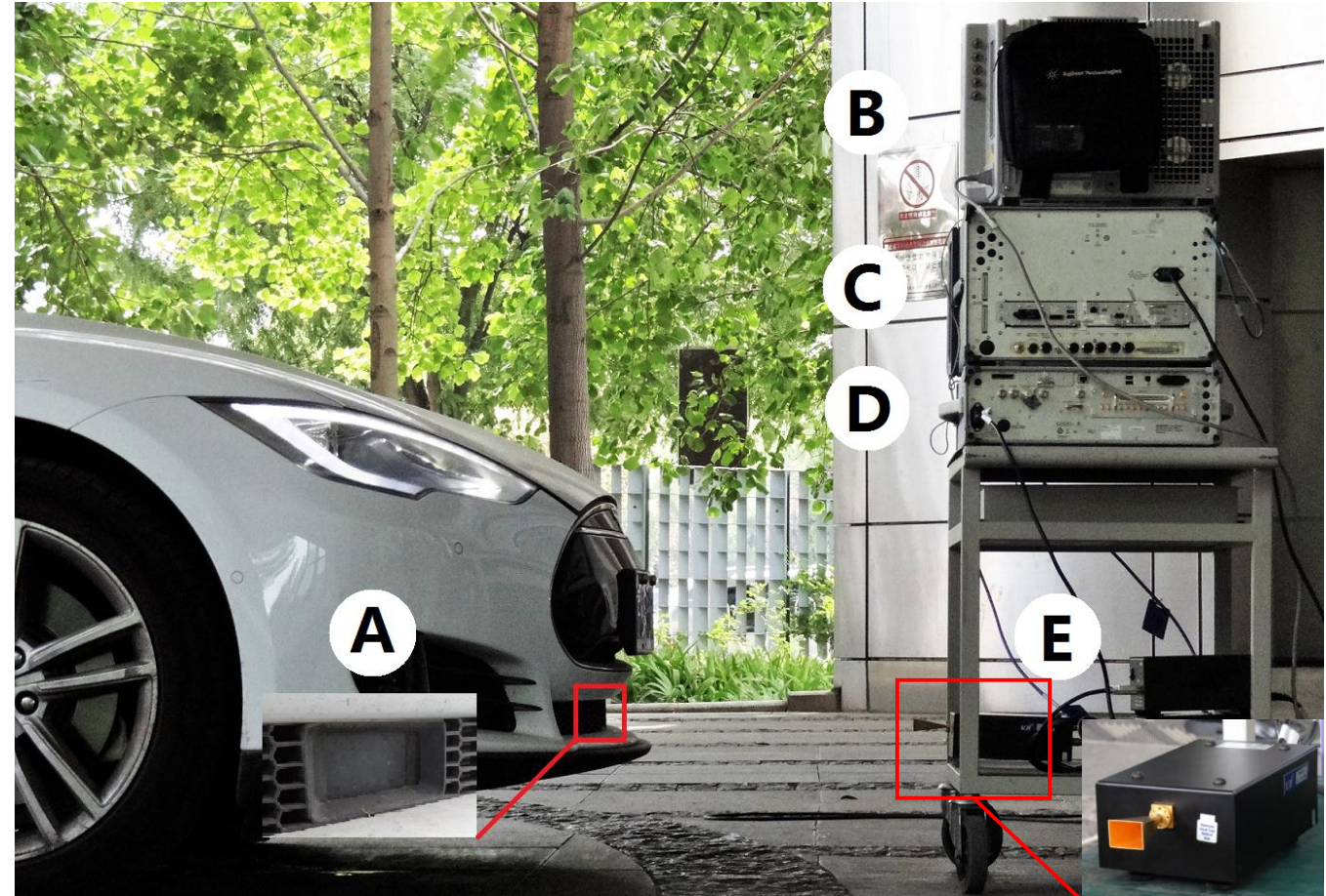


The MMW Radar on Tesla Model S

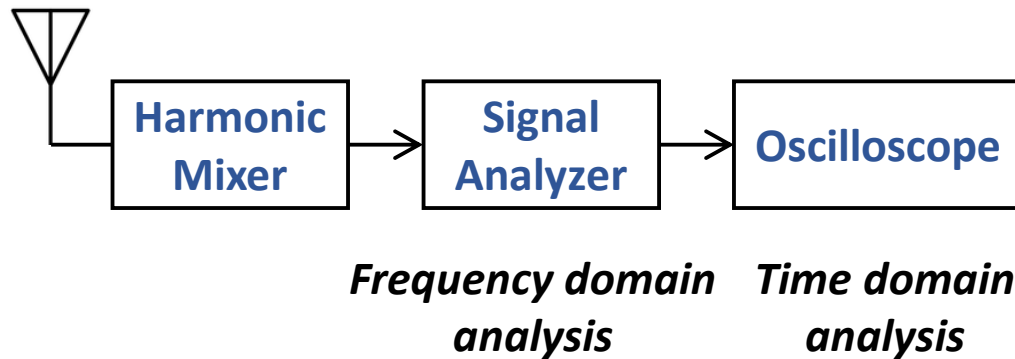
Attacking MMW Radar – Setup

- **Signal Analysis**
- **Jamming Attack**
- **Spoofing Attack**

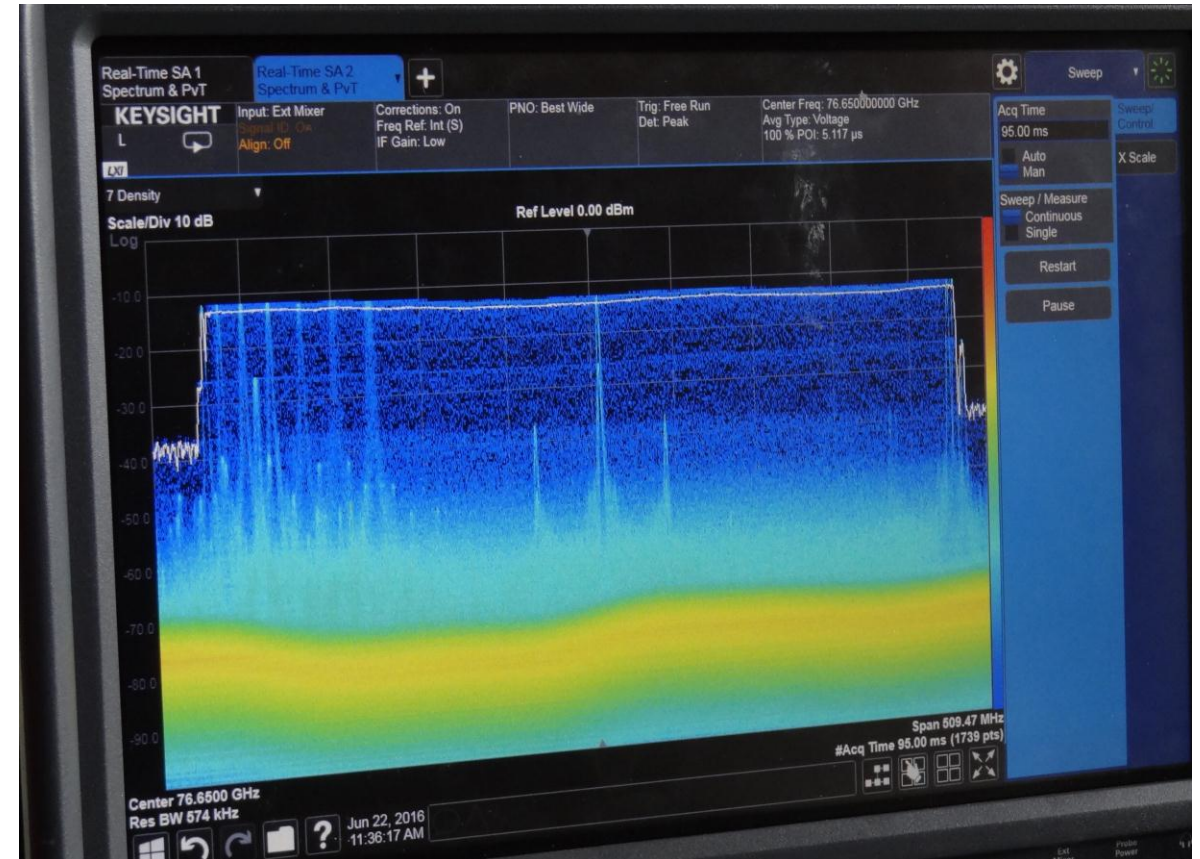
- **Equipment:**  **KEYSIGHT TECHNOLOGIES**
 - Tesla Model S Radar (A)
 - Signal analyzer (C)
 - Harmonic mixer (E)
 - Oscilloscope (B)
 - Signal generator (D)
 - Frequency multiplier (E)



MMW Radar Signal Analysis



- **Center frequency: 76.65 GHz**
- **Bandwidth: 450 MHz**
- **Modulation: FMCW**
- **Radar chirp details ...**



Real-time spectrum on signal analyzer

What indicates Autopilot?

- **What does blue mean?**
- **Why stationary?**

Traffic Aware Cruise Control is on.

Autosteer is on.



Jamming Attack – Demo



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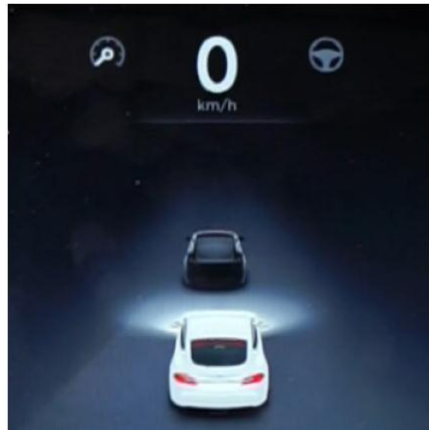
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Jamming hides obstacles.

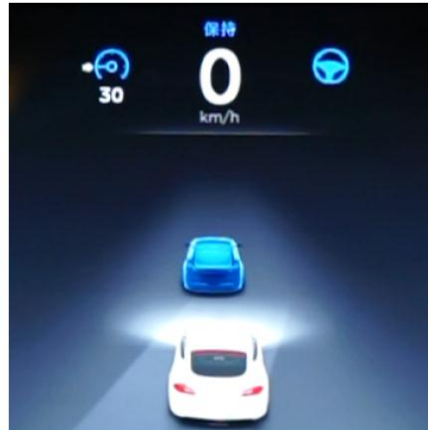


Attacking MMW Radars – Results

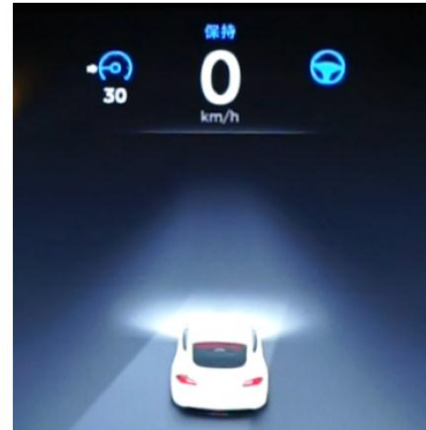
- **Jamming:** **hides** detected objects
 - Either fixed or sweeping frequency signal worked
- **Spoofing:** **alters** object distance



(a) Drive gear.



(b) Autopilot.



(c) Jammed.

Result of jamming attack

Attacking Cameras

Mobileye & Point Grey

Tesla Model S

Automotive Cameras

What is automotive camera?

- **Computer vision**
- **Forward & backward**

- **Applications**

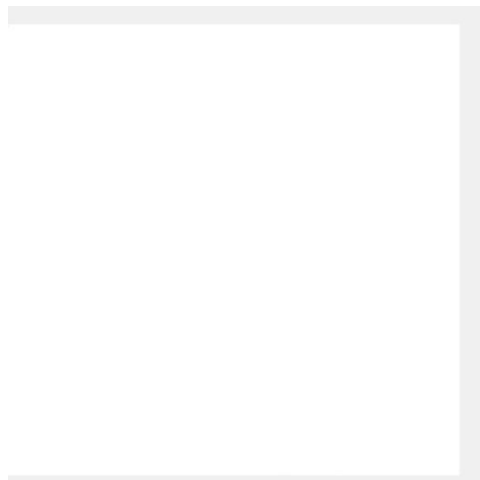
- Lane departure warning
- Lane keeping
- Traffic sign recognition
- Parking assistance



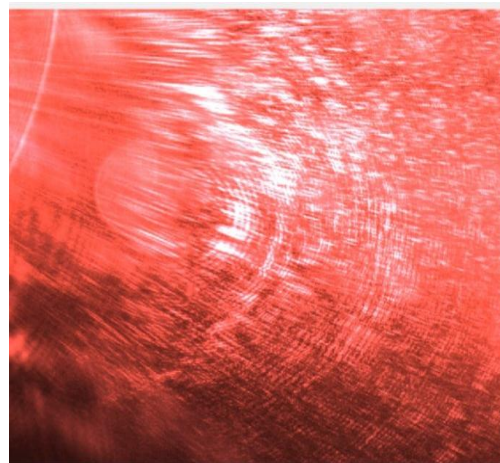
Blinding Cameras – Results with Laser beam

Total blinding

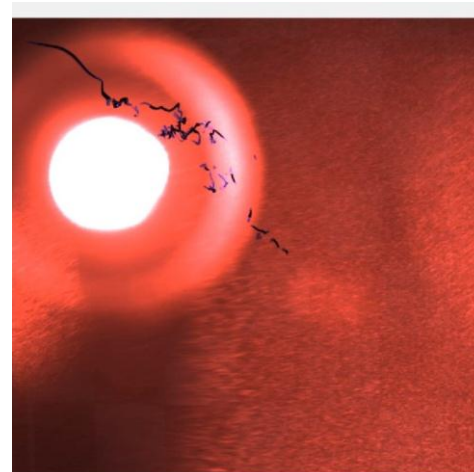
Total blinding



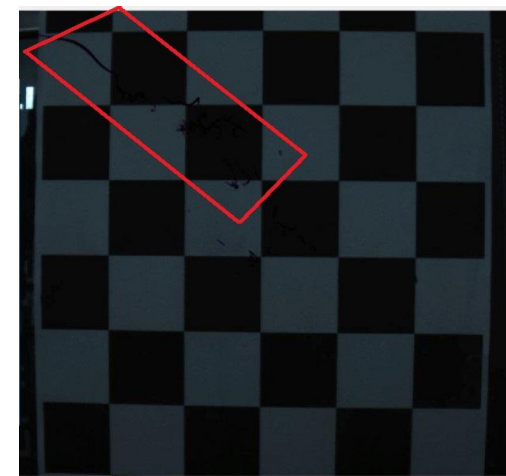
Fixed laser beam



Wobbling laser beam



Damaged



Permanently damaged

Blinding Cameras – Demo with Laser beam



Response from Tesla



“... We appreciate the hard work you have put into researching potential attacks on sensors used in the Autopilot system. We are currently evaluating your report and investigating the concerns your team has raised so that we can understand if any real world risks have been uncovered ...”

Countermeasures

- **Sensor fail safe**
 - Zero or maximum
 - Anomaly detection
- **Sensor redundancy**
 - MIMO system
 - Different types of sensors
- **Sensor data fusion**



Conclusions and Takeaway messages

- **Attacking existing sensors is **feasible****
- **The sky is not falling**
- **Sensors should be designed with security in mind**
 - Think about **intentional attacks**
- **For customers**
 - Don't trust **semi-autonomous** cars yet

Will we have fully secure autonomous cars?



Acknowledgements



- **Tongji University**

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KEYSIGHT
TECHNOLOGIES

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- **USSLab, Zhejiang University**

- Weibin Jia, Zhou Zhuang, Guoming Zhang

- **ADLAB, Qihoo 360**

- Bin Guo





Questions and Answers

Check out our whitepaper!

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谢谢



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